Chuqi Wang

+44 7421 265183
wangchuqi@gmail.com

Education

- **IMPERIAL COLLEGE LONDON**
  MSc Biomedical Engineering, Neurotechnology Stream 2022 – 2023
- **Relevant Modules:** Neuroscience, Brain Machine Interfaces, Computational Neuroscience, Mathematical Methods for Neural Science and Engineering, etc.

- **UNIVERSITY COLLEGE LONDON**
  BSc Physics with Medical Physics 2019 – 2022
- **Relevant Modules:** Mathematical Methods, Computing in Medicine, MRI and Biomedical Optics, Medical Electronics and Neural Engineering, Introduction to Biophysics, etc.

- **QUEEN ANNE'S SCHOOL** Reading
  A-levels 2017 – 2019
  Mathematics (A), Physics (A), Chemistry (A)

- **READING GIRLS' SCHOOL** Reading
  GCSEs 2015 - 2017
  11 GCSEs, Grade A* - A in maths and all sciences

Projects

02/2023-10/2023 Tracking Changes in Brain Organoid Spontaneous Activity Over Maturation

Summary: Developed a temperature control system for image recording of the brain organoid. Collecting and analysing of calcium imaging data of early age human and mouse cerebral organoid maturity, neural activity and network-level dynamics.

- Utilised a two-photon microscope to capture high resolution images of biological samples
- Integrated a feedback loop circuit to enable proportional temperature control
- Designed the mechanical layout of an imaging chamber and refined the electronic connections to maximise stability throughout the data collection process

01/08/2022-23/09/2022 Characterisation of a Multi-User Ultrasound Therapy System
Research Assistant, Biomedical Ultrasound Group UCL

Summary: This project revolved around the examination of a ultrasound therapy system with new launched GUI software. Conducted calibration experiments and collaborated with the supplier company to compose a user guide of the system.

- Efficiently managed multiple tasks, ensuring timely completion and making necessary adjustments to the experimental design as required
- Employed critical thinking skills to address challenges encountered in testing procedure, ultimately validating the experimental results
- Developed a system guideline as a reference for interdisciplinary researchers, outlining parameter settings and troubleshooting guide

01/2022-06/2022 Biofeedback with Electromyography Sonification and Music

Summary: Built a prototype that converts muscle signals to musical sound and test its possibilities in biofeedback. Experienced in signal processing, electronics, understanding auditory nervous system, and tackle engineering problem.

- Designed a signal filtering electrical circuit with a focus on enhancing signal resolution and accuracy, optimising overall signal performance
- Utilised MATLAB for data processing and Arduino programming to develop a real-time conversion prototype
- Successfully capture forearm electromyography (EMG) signal and converted them into audible tone

Extra-curricular Activities

10/2019-06/2022 UCL Chinese Student and Scholars Association
Executive

Teamwork:

- Contributed to the self-organised performances for events and festivals
- Played a role in promoting events through marketing efforts

2018-2022 The Duke of Edinburgh's Gold Award

Resilience:

- Undertook a challenging four-day hiking expedition in the Lake District
- Stepped out of my comfort zone, gaining valuable experience in decision-making as part of a team

Summer 2017 National Citizen Service, Reading

Project management:

- Engaged with a student team to coordinate a group activity aimed at enhancing the wellbeing of residents in a care home
- Contributed to writing the project plan, actively participated in fundraising and campaign activities in the event's preparation phase

Skills

- Wet lab, Microscopy, Electronics and Ultrasound
- Programming: MATLAB, Python
- Data Collection and Analysis
- Research & Development
- Report Writing
- Language: English, Chinese

References

References available upon request